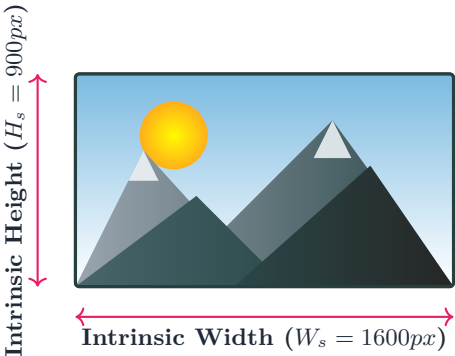


Algorithmic Engine of object-fit: contain

Evaluating Disparate Dimensional Variables & Boundary Calculations

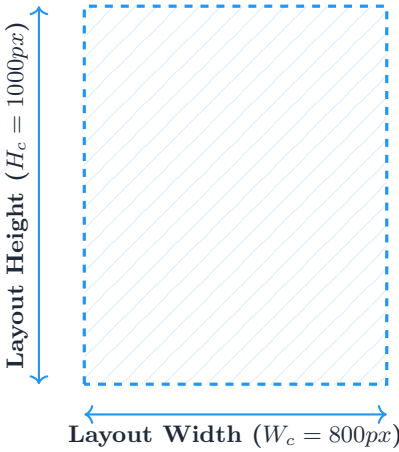
SOURCE MEDIA (X_s, Y_s)



Aspect Ratio (R_s): $W_s \div H_s = 1.77$ (16:9)

The media's raw, unconstrained pixel dimensions mapped directly from the file buffer.

CSS LAYOUT CONTAINER (X_c, Y_c)

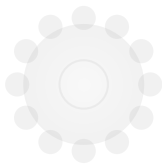


Aspect Ratio (R_c): $W_c \div H_c = 0.8$ (4:5)

The rigid DOM bounding box computed by the CSS engine layout algorithm.

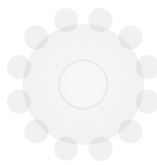
DIMENSIONAL EVALUATION ENGINE

Step 1: Calculate Axis Scale Constraints



$$S_x = \frac{W_c}{W_s} = \frac{800}{1600} = \mathbf{0.50}$$

$$S_y = \frac{H_c}{H_s} = \frac{1000}{900} = \mathbf{1.11}$$

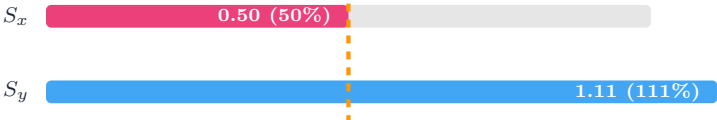


The engine computes the required scaling multipliers to individually span the X and Y layout bounds.

THE INTERPOLATION HALT

Step 2: The Collision Threshold

Limit (0.50) [HALT]



Engine Logic: $S_{\text{final}} = \min(S_x, S_y)$

The rendering engine expands the matrix uniformly. It evaluates the bounds and explicitly **halts interpolation** the moment the lowest boundary scale is met.

Here, $\min(0.50, 1.11) = 0.50$. The X-axis boundary is struck first.

triggers

FINAL RENDERED OUTPUT

Step 3: Computed Box Dimensions

Final Width: $W_{out} = W_s \times S_{final} = 1600 \times 0.5 = \mathbf{800px}$

Final Height: $H_{out} = H_s \times S_{final} = 900 \times 0.5 = \mathbf{450px}$

The object perfectly spans the container's width (800px), generating a residual void of $1000 - 450 = 550px$ on the Y-axis.

VISUAL RESULT (LETTERBOXING)

